

KARTHIK TATINENI

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EDUCATION

B.Tech in Electronics and Communication Engineering

2022 – 2026

Institute of Aeronautical Engineering, Hyderabad | CGPA: 7/10

SKILLS

Languages: Python, SQL

AI/ML: PyTorch, LLMs, Transformers, RAG, Embeddings, YOLO, CNN, AI Agents, MCP, LangChain

Frameworks: FastAPI, Next.js

Databases: PostgreSQL, pgvector, MongoDB, MySQL, Redis

Cloud: AWS(EC2, S3), Azure (Container Apps, AI-Foundry), Docker, Cloudflare, Grafana

Tools: PowerBI, Git, Postman, Hugging Face, Ollama, NVIDIA NIM, JupyterLab, Anaconda, Pandas, Claude Code CLI, Google SEO

EXPERIENCE

Freelance Software Developer

Jan 2025 – Present

- Built **KLVORA Fashion Hub**, a production-ready Next.js e-commerce platform with Supabase, AI support, Three.js, Google SEO integration, secure authentication, inventory management, and order tracking. [[Live](#)]
- Built **Touchwood Interiors CMS**, a production-ready Next.js platform with Sanity CMS, Supabase, AI support chatbot, Google SEO integration, secure quotations, interactive 3D visualization, and architect portfolios. [[Live](#)]

PROJECTS

DisasterMind AI - Multi-Agent | *FastAPI, NVIDIA NIM, PostgreSQL, pgvector, RAG, Embedding* | [[GitHub](#)] [[Live Demo](#)]

- Built a multi-agent RAG system using NVIDIA NIM LLMs, embeddings, and pgvector similarity search over PostgreSQL to power explainable disaster response reasoning across coordinator, risk, logistics, evacuation agents.
- Engineered an async FastAPI backend with tool-calling agent orchestration, retrieving 12,000+ real-time records via RAG to reduce hallucination and power a Next.js/TypeScript decision-support dashboard.

Aegis AI - Local-First AI Assistant Platform | *FastAPI, PyTorch, Transformers, ChromaDB, RAG* | [[GitHub](#)] [[Live Demo](#)]

- Built a local-first AI assistant running open-source LLMs (Mistral 7B, Qwen 2.5) using PyTorch and Transformers, deployed on a Lightning AI GPU instance (NVIDIA T4) for inference.
- Implemented a RAG pipeline with ChromaDB vector storage to give the assistant long-term memory, letting it retrieve and reference earlier conversations to answer follow-up questions.

Real-Time Helmet & Triple-Ride Detection | *Python, YOLO, CNN, TensorFlow, OpenCV, Tkinter* | [[GitHub](#)] [[Live Demo](#)]

- Built a computer vision pipeline with custom-trained YOLO and CNN models for helmet and number-plate detection, deployed as a web app for live image-upload inference.
- Tested the pipeline on sample traffic images and manually verified detections, then automated violation handling with Excel logging and email alerts carrying the recognized plate number.

PUBLICATIONS

T. Karthik Sai et al., **Helmet and Triple-Ride Detection Using YOLO and CNN**, ICMCER 2026 (Scopus Indexed) | [Certificate](#)

CERTIFICATIONS

- Microsoft Certified(AI-901): Azure AI Fundamentals | [Certificate](#)
- Microsoft Certified(AI-103): Azure AI Apps and Agents Developer Associate | [Certificate](#)
- Microsoft AI Skills Fest 2026 | [Credly Badge](#)
- MySQL & Database Management: Create, Manage & Query Databases | [Certificate](#)
- HTML & CSS: The Complete Web Development Guide | [Certificate](#)

EXTRA-CURRICULAR

- Hosted and organised the official **IARE Consortium** event for 1,000+ participants; designed and developed the end-to-end registration platform.
- **Core Member and Logistics Head**, NSS – led community logistics for college and district level programs.